

AKSHAY BAIPLAWAT

IMPACT DRIVEN SOFTWARE ENGINEER WITH A PASSION FOR PROBLEM SOLVING

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Education

MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD , India

2019 - 2023

BACHELOR OF TECHNOLOGY | CPI: 7.02 / 10

Skills

AI & Agentic Development:

- **LLM Orchestration:** Multi-agent pipeline development utilizing Gemini and Claude.
- **AI Engineering:** Proficiency in Claude Code, Prompt Engineering, RAG (Retrieval-Augmented Generation) systems, semantic search via Mem0, and multimodal context injection.

Core Backend & Systems

- **Languages:** Expert-level C++, Core Java, JavaScript, Groovy, Apex and SQL.
- **Frameworks:** Spring Boot, React, Kibana (OctoBus) and SAP CPI.
- **Fundamentals:** Advanced Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), and DBMS.
- **Systems Programming:** Memory management (C++), multi-threading, and Windows Native (Win32) development.
- **Tools:** Claude Code, CI/CD (GitHub Actions) , Git, GitHub, VS code, Visual studio (Windows Native) and CMake.

Work Experience

SAP LABS INDIA (Concur Integration) | Developer Associate

Nov' 24 - Present

- **Architected a proactive scalability service using custom Mock APIs** to simulate extreme data loads, identifying critical **JMS Queue and memory bottlenecks** that would have blocked onboarding for **1,000+ future enterprise customers**.
- **Reduced artifact transport time by 99% (from 5 minutes to seconds)** by architecting an automated transport tool for internal engineering teams, effectively saving 50+ manual engineering hours per month.
- **Lead Technical Point of Contact (KPC) for the Salesforce Connector V2 migration**, managing the end-to-end upgrade of legacy APIs and resolving customer-facing defects (CRMCs) to ensure a seamless transition before the Q3 deadline.
- **Designated Product Line Owner for the ADP line (WorkforceNow, WorkforceNow NextGen, Run, Vantage)**, orchestrating high-scale service updates and maintaining **100% alignment with release calendars** to guarantee zero-downtime for global enterprise clients.
- **Optimized system observability as Incident Manager (2025-2026 Jan)**, overhauling PagerDuty alert logic and OctoBus logging dashboards to accelerate "On-Call" response times and production story resolution.
- **Executed a dual-track Development and Quality Engineering strategy**, personally uncovering and neutralizing **two critical production defects** that threatened the entire **100% Concur Professional Entity NetSuite user base**.

ZOHO Corporation | Software Developer

Aug' 23 - Nov' 24

- **Engineered and maintained the Zoho Assist Windows Agent**, ensuring **100% compatibility** across all Windows OS versions for a global remote access user base
- **Optimized the session initialization flow**, achieving a **30% reduction in latency** and a **15% decrease in executable size**, improving the experience for **10 Lakh daily sessions**.
- **Implemented a session audit reporting feature** that enhanced customer security metrics by **35%** through detailed session logging and transparency.

- **Neutralized a critical, industry-wide architectural bug “block touch pad cursor”**, positioning Zoho Assist as the first remote support software to operate without the defect.
- **Collaborated with cross-functional teams**, resolving integration issues and helping them successfully integrate Zoho Assist into their products.
- **Streamlined third-party library integration (POCO, rapidJSON)** by developing custom wrappers, resulting in a **15% performance gain** in core agent functions.
- **Collaborated with cross-functional teams** to resolve complex integration issues, facilitating the successful embedding of Zoho Assist into multiple partner product lines.

Projects

Finance Tracker | UI, Engineer & Architect (Utilizing Gemini and Claude) [Link](#)

- **Architected** a privacy-centric Android application using **MVVM and Clean Architecture** (Domain/Data/Presentation) with **Jetpack Compose** and **Kotlin Flow**, ensuring 100% reactive data streams and zero-cloud-dependency for GDPR/DPDP compliance.
- **Engineered** an intelligent **SMS Parsing Engine** with regex-based pattern matching and merchant normalization to automate transaction categorization for major Indian banks (HDFC, ICICI, SBI), eliminating manual entry.
- **Optimized** group debt resolution by implementing a **Greedy Settlement Algorithm**, reducing potential transaction complexity from **$O(N^2)$ to $O(N)$** and validating logic through 23+ targeted unit tests.
- **Orchestrated** complex background synchronization using **WorkManager**, handling periodic SMS syncing, subscription reminders, and automated weekly debt summaries with persistence across device reboots.
- **Achieved** high system reliability and code quality by authoring **180+ unit tests** across 13 test suites, covering critical edge cases like OTP exclusion, duplicate prevention, and CSV formula injection protection.
- **Developed** custom UI components including a keyboard-aware bottom sheet to bypass **Material3** limitations, improving the data-entry experience across 11+ navigable screens.

JARVIS: Multimodal AI Voice Assistant | AI Engineer [Link](#)

- **Architected** a real-time multimodal agent using **LiveKit Agents SDK** and **WebRTC**, orchestrating a low-latency STT → LLM → TTS pipeline with **<1s perceived response time** through Voice Activity Detection (VAD) and asynchronous thinking-state feedback.
- **Engineered** a custom **VisionAssistant** class to process asynchronous video streams, automatically injecting real-time visual context into **Gemini 2.5 Flash** for vision-grounded reasoning and situational awareness.
- **Integrated** a persistent memory layer using **Mem0** and **vector similarity search**, enabling cross-session personalization by loading semantic context and user preferences at session initialization.
- **Orchestrated** complex productivity workflows via the **Model Context Protocol (MCP)**, allowing the LLM to autonomously manage **Google Calendar** operations and **Gmail SMTP** dispatch through secure tool-calling patterns.
- **Developed** a production-grade tool suite with defensive error handling for 6+ external APIs (Deepgram, ElevenLabs, Perplexity), managing **8+ exception types** including rate limits and authentication failures to ensure 99% session uptime.
- **Implemented** a **Test-Driven Development (TDD)** framework utilizing an **LLM-as-judge** evaluation pattern to validate grounding, intent-based behavior, and safety protocols across 1,000+ lines of Python code.

Professional Engineering Portfolio | UI, Architect & Engineer (Utilizing Gemini and Claude) [Link](#)

- **Architected** a high-performance platform using **TypeScript 5.4** and **Vite**, implementing a **Singleton-based OOP architecture** and **Plugin Facade pattern** to manage 7+ integrations with explicit memory-leak prevention.
- **Engineered** a **4-stage parallelized CI/CD pipeline** via GitHub Actions, reducing deployment time by **40%** while automating production-grade security audits via **Trivy vulnerability scanning**.
- **Optimized** system performance to achieve **Lighthouse scores of 95+** and **SEO excellence**, utilizing **Vitest** for **>80% code coverage** and maintaining a **<1.5s First Contentful Paint (FCP)**.